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7

COLLISIONLESS FLOWS

7.1 Bimodal distributions

As discussed in Chapter 1, collisionless or free-molecule flows are the limiting case in which an appropriately defined Knudsen number tends to infinity, i.e.,

$$(Kn) \equiv \lambda / L \rightarrow \infty . \tag{7.1}$$

Collisionless flows are associated either with very low densities, and therefore with very high mean free paths, or with very small typical dimensions.

In the absence of intermolecular collisions, collisionless flowfields are formed by the superimposition of several classes of molecule. These classes could be separate streams with different temperatures and with a relative velocity between them, or a freestream and the reflected molecules from a surface. In most cases, these separate classes will each conform to an equilibrium or Maxwellian distribution. The superimposition may be such that the whole or only part of the distributions contribute to the region of interest in the collisionless flow.

First consider the case in which two *full-range* distributions contribute to the flow. These will be denoted by the subscripts 1 and 2. They are at separate temperatures and, without loss of generality, the relative velocity of magnitude c_r may be chosen to be along the x-axis of gas 2. The gas then comprises a number density n_1 of stationary gas defined by

$$f_1 = (\beta_1^3 / \pi^{3/2}) \exp(-\beta_1^2 \{u^2 + v^2 + w^2\}) \tag{7.2}$$

and n_2 of gas with a stream speed c_r and a distribution function

$$f_2 = (\beta_2^3 / \pi^{3/2}) \exp(-\beta_2^2 \{[u - c_r]^2 + v^2 + w^2\}) . \tag{7.3}$$

The combined distribution function is

$$nf = n_1 f_1 + n_2 f_2 ,$$

and the macroscopic properties or moments of the distribution function are given by eqn (3.3) as

$$\bar{Q} = \frac{1}{n} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (n_1 Q_1 f_1 + n_2 Q_2 f_2) \, du \, dv \, dw . \tag{7.4}$$

The mean molecular mass is obtained by setting $Q = m$, to give

$$\bar{m} = (n_1 m_1 + n_2 m_2)/n. \tag{7.5}$$

The stream speed of the mixture is defined by eqn (1.43) and is obtained by setting $Q = mu$ and then using eqn (7.5), to give

$$c_0 = \frac{n_2 m_2}{n_1 m_1 + n_2 m_2} c_r. \tag{7.6}$$

The translational temperature is best obtained through the alternative definition of eqn (1.51) by setting $Q = mc^2$ and then using eqns (7.5) and (7.6), to give

$$T = \frac{1}{n}\left(n_1 T_1 + n_2 T_2 + \frac{n_1 m_1 n_2 m_2}{m_1 n_1 + m_2 n_2}\frac{c_r^2}{3k}\right). \tag{7.7}$$

This is most instructively written as

$$T = \bar{T}\left(1 + \frac{2n_1 n_2}{3n^2}\bar{s}^2\right), \tag{7.8}$$

where

$$\bar{T} = (n_1 T_1 + n_2 T_2)/n$$

and

$$\bar{s} = c_r/(2k\bar{T}/\bar{m})^{1/2} = c_r\bar{\beta}.$$

The superimposition of two cool gas streams with a high relative velocity can therefore lead to high kinetic temperatures, even though there are no collisions between the molecules of the two streams. This is because the combination of two equilibrium velocity distributions with a significant relative velocity between them leads to a highly non-equilibrium velocity distribution function. The kinetic temperature is properly defined for all distribution functions, whereas the thermodynamic temperature has no meaning in other than an equilibrium situation (when it is equal to the kinetic temperature). Kinetic temperatures in collisionless flows must not be interpreted as, or confused with, thermodynamic temperatures.

In most cases, the geometry is such that only part of each distribution function applies at a particular location. For example, consider the case in which a stream of density n_∞, speed c_∞, and temperature T_∞ is directed at an angle θ to the normal to a solid surface. The coordinate system is the same as that adopted for the molecular flux calculations in §4.2, and was illustrated in Fig. 4.2. The distribution function for the stream is

$$f_\infty = (\beta_\infty^3/\pi^{3/2})\exp(-\beta_\infty^2\{[u - c_\infty\cos\theta]^2 + [v - c_\infty\sin\theta]^2 + w^2\}), \tag{7.9}$$

and, if it is assumed that the surface interaction is diffuse with complete accommodation to the surface temperature T_w, the distribution function for the reflected molecules is

$$f_r = (\beta_w^3/\pi^{3/2})\exp(-\beta_w^2\{u^2 + v^2 + w^2\}). \tag{7.10}$$

The number density n_r that is associated with the reflected molecules is given by the condition that, in a steady state, the flux of molecules from the surface must be equal to that to the surface. Therefore, from eqns (4.22) and (4.24)

$$\frac{n_r}{2\pi^{1/2}\beta_w} = \frac{n_\infty}{2\pi^{1/2}\beta_\infty}(\exp\{-s^2\cos^2\theta\} + \pi^{1/2}s\cos\theta\{1 + \mathrm{erf}[s\cos\theta]\}),$$

so that

$$n_r = n_\infty(T_\infty/T_w)^{1/2}(\exp\{-s^2\cos^2\theta\} + \pi^{1/2}s\cos\theta\{1 + \mathrm{erf}[s\cos\theta]\}), \quad (7.11)$$

where

$$s = c_\infty\beta_\infty = c_\infty/(2kT_\infty/m)^{1/2}.$$

The gas is homogeneous in composition and m applies to both the incident and reflected molecules. The moment equation must now be based on two *half-range* distributions and may be written

$$\bar{Q} = \frac{1}{n}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(\int_0^{\infty} n_\infty Q_\infty f_\infty\,\mathrm{d}u + \int_{-\infty}^{0} n_r Q_r f_r)\,\mathrm{d}u\right)\mathrm{d}v\,\mathrm{d}w. \quad (7.12)$$

The number density n above the surface may be obtained by evaluating this equation with the Q's set equal to unity. This gives

$$n = \{1 + \mathrm{erf}(s\cos\theta)\}n_\infty/2 + n_r/2,$$

or, using eqn (7.11),

$$n = \frac{n_\infty}{2}\Big(1 + \mathrm{erf}(s\cos\theta) + (T_\infty/T_w)^{1/2}(\exp\{-s^2\cos^2\theta\} + \pi^{1/2}s\cos\theta\{1 + \mathrm{erf}[s\cos\theta]\})\Big), \quad (7.13)$$

This density applies at the surface and as far above the surface as the flow remains macroscopically uniform. When $s\cos\theta$ is above about 3, the exponential term becomes negligible and the error function is very close to unity. The density can then be written

$$n = n_\infty\{1 + (T_\infty/T_w)^{1/2}\pi^{1/2}s\cos\theta\}. \quad (7.14)$$

The components of the stream velocity can be evaluated by setting Q to the corresponding molecular velocity components. The velocity component in the x-direction is normal to the surface and, as expected, it is zero for the combined distribution. The velocity component of the combined distributions parallel to the surface in the y-direction is

$$v_c = \tfrac{1}{2}(n_\infty/n_r)c_0\sin\theta(1 + \mathrm{erf}\{s\sin\theta\}). \quad (7.15)$$

Eqn (7.11) can again be substituted for n_r, but the origins of the *slip velocity* are clearer if this is not done. The *temperature jump* can also be calculated from eqn (7.12), but the expression is excessively complicated.

7.2 Molecular effusion and transpiration

Molecular effusion is the term that is applied to the flow that occurs when an equilibrium gas is separated from a vacuum by a thin wall in which there is a small hole. If the dimensions of the hole are sufficiently small in comparison with the mean free path, each molecule that passes through the hole is subject to a negligibly small probability of being affected by the flow of other molecules towards or through the hole. The flux through the hole is then the same as that across any small surface element in an equilibrium gas, as analysed in § 4.2. The number flux is given by eqn (4.24) and, multiplying this by m, the free-molecule mass flux per unit area is

$$\Gamma_f = \frac{nm}{2\pi^{1/2}\beta} = \rho\left(\frac{RT}{2\pi}\right)^{1/2}. \tag{7.16}$$

The free-molecule flux may be compared with the result from continuum theory which is valid when the mean free path is extremely small in comparison with the dimensions of the hole. The inviscid continuum model is a choked flow with sonic velocity in the plane of the hole. The continuum mass flux is, therefore,

$$\Gamma_c = \rho^* a^*$$

with the asterisk representing sonic conditions. The macroscopic quantities in the free-molecule result are the stagnation conditions and, using the standard one-dimensional steady continuum flow relationships between the sonic and stagnation conditions,

$$\Gamma_c = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \rho\,(\gamma RT)^{1/2}. \tag{7.17}$$

Here, γ is the specific heat ratio of the gas so that the continuum and free-molecule results differ only by a numerical factor.

A comparison of eqns (7.16) and (7.17) shows that the ratio of the free-molecule to the continuum mass flux is

$$\frac{\Gamma_f}{\Gamma_c} = \frac{1}{(2\pi\gamma)^{1/2}}\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} = 0.5494 \quad \text{for } \gamma = 5/3\,. \tag{7.18}$$

Assuming a circular hole, the obvious choice of Knudsen number is the ratio of the mean free path in the undisturbed gas to the diameter of the hole. For a problem with a very simple geometry, such as this, it would be expected that the transition between the two limits would be monotonic with the major change occurring between Knudsen numbers of 0.1 and 10. Liepmann (1961) has discussed the effusion flow in depth and reports experiments that generally confirm these expectations.

The interaction of separate molecular streams and molecular species in gas flows takes place through intermolecular collisions and, since these are absent in collisionless flows, a relatively complex flow may be built up by the superimposition of simpler flows. Several important examples of this involve combinations of effusion flows.

First, consider the case in which there is an equilibrium gas on both sides of the thin wall containing the small orifice. For sufficiently high Knudsen number, effusion will occur in both directions with each stream of gas being quite independent of the other. If the gas on one side of the orifice is contained in a vessel that is maintained at temperature T_A, and that on the other side is in a vessel maintained at temperature T_B, there must be an equal flux of gas in each direction. If there is a similar simple gas in both vessels, eqn (7.16) gives

$$\rho_A (RT_A)^{1/2} = \rho_B (RT_B)^{1/2},$$

and, using eqn (1.28),

$$\frac{p_A}{p_B} = \left(\frac{T_A}{T_B}\right)^{1/2}, \tag{7.19}$$

so that a pressure difference is established between the two vessels. This phenomenon is called *thermal transpiration.*

Next consider the case in which a mixture of species 1 and species 2 molecules is separated from a vacuum by a membrane that contains a number of free-molecule orifices. After passing through the membrane, the gas is collected. The number density ratio in the collected gas will be proportional to the number flux ratio and, again using eqn (7.16) and noting that $R = k/m$,

$$\frac{n_1'}{n_2'} = \frac{n_1}{n_2}\left(\frac{m_2}{m_1}\right)^{1/2}. \tag{7.20}$$

The prime denotes the collected gas, and eqn (7.20) shows that there is an enhancement in the number density of the light gas.

The spatial distribution of the gas in the effusion flowfield is also of interest. The number density of the gas upstream of the orifice will now be denoted by n_0 to distinguish it from the number density n in the effusing gas. The number of molecules per unit time effusing from an orifice of area S that have a speed between c and $c + dc$, and that also make an angle between θ and $\theta + d\theta$ with the normal to the plane of the orifice, is given by eqn (4.39) as

$$2\pi^{-1/2} n_0 S \beta^3 c^3 \exp(-\beta^2 c^2) \sin\theta \cos\theta \, d\theta \, dc.$$

When these molecules have moved a distance r (that is large in comparison with the radius of the orifice) they cross an annulus of area $2\pi r^2 \cos\theta d\theta$, so that the number per unit time is also given by

$$2\pi dn c r^2 \cos\theta d\theta.$$

The number density dn due the class c molecules is therefore

$$\mathrm{d}n = \frac{n_0 S \cos\theta}{r^2} \frac{\beta^3}{\pi^{3/2}} c^2 \exp(-\beta^2 c^2)\mathrm{d}c$$

and, integrating over all speeds from 0 to ∞,

$$n = \frac{n_0 S \cos\theta}{4\pi r^2}. \tag{7.21}$$

There is therefore a 'cosine distribution' superimposed on an 'inverse square' decay of the number density in the effusing molecules. While eqn (7.21) requires that r should be large compared with the radius r_0 of the orifice, it may be used to obtain an exact expression for the number density along the normal from the centre (i.e. the axis or centreline of the effusing flow). Consider an annular element of width dx at a radius x in the plane of the orifice. With respect to this element, the axis is such that $\cos\theta = r/(r^2+x^2)^{1/2}$ and the number density along the axis is

$$n = \frac{n_0 r}{2} \int_0^{r_0} \frac{x\,\mathrm{d}x}{(x^2+r^2)^{1/2}}$$

or

$$n = (n_0/2)\,[1 - \{1 + (r_0/r)^2\}^{-1/2}]. \tag{7.22}$$

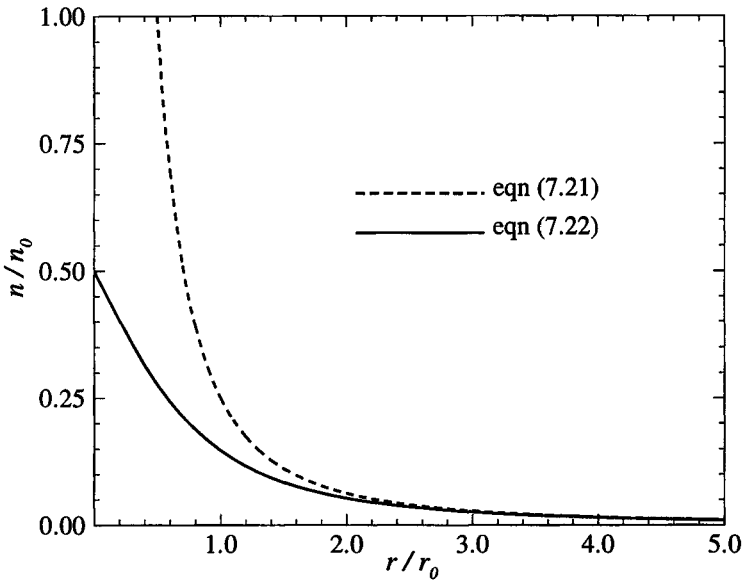


Fig. 7.1 Number density along the normal from the centre of an orifice.

7.3 One-dimensional steady flows

A flow may be regarded as one-dimensional if it can be solved through the application of the molecular flux equations in one direction. The mass flux aspect of the molecular effusion problem qualifies as one-dimensional, even though the flowfield is two or three-dimensional. This is the simplest class of flows and the free-molecule limits of the classical Couette flow and the parallel plate heat transfer problem may be compared with the corresponding continuum flows.

Consider the *one-dimensional heat transfer* between two infinite plane parallel plates separated by a distance h that is very small in comparison with the mean free path λ of the gas between the plates. Assume diffuse reflection with complete thermal accommodation to the temperature T_U at the upper plate, and T_L at the lower plate. The monatomic gas of number density n between the plates then consists of an upward moving stream of density n_L and temperature T_L emitted from the lower plate, together with a downward moving stream of density n_U and temperature T_U emitted from the upper plate.

The flux of diffusely reflected molecules from a surface is identical with the effusion of molecules from a fictitious equilibrium gas on the reverse side of the surface. The number density of the reflected molecules is half that of the fictitious gas, since half of these are moving in the direction normal to the surface. Therefore, from eqns (4.23) and (4.24),

$$n_U T_U^{1/2} = n_L T_L^{1/2}$$

and, since

$$n = n_U + n_L ,$$

$$n_L = \frac{n T_U^{1/2}}{T_U^{1/2} + t_L^{1/2}} , \qquad n_U = \frac{n T_L^{1/2}}{T_U^{1/2} + T_L^{1/2}} . \tag{7.23}$$

The energy flux q_L from the lower plate is obtained by setting $\beta = (2RT_L)^{-1/2}$ and $\rho = 2n_L m$ in eqn (4.27), to give

$$q_L = m n_L \pi^{-1/2} (2RT_L)^{3/2} .$$

Similarly, the downward energy flux from the upper plate is

$$q_U = m n_U \pi^{-1/2} (2RT_U)^{3/2} .$$

The net upward heat flux is obtained by combining these equations with eqn (7.23), to give

$$q_f = -2^{3/2} \rho \pi^{-1/2} R^{3/2} T_U^{1/2} T_L^{1/2} (T_U^{1/2} - T_L^{1/2}) . \tag{7.24}$$

The continuum heat transfer between the plates is

$$q_c = -K(\mathrm{d}T/\mathrm{d}y) ,$$

with the y-axis normal to the lower plate. If we assume that $K = CT^{\omega}$ with C and η constant, this becomes

$$q_c = -\frac{C}{\omega+1}\frac{dT^{\omega+1}}{dy}.$$

The continuum energy equation in a static gas requires that q_c be a constant, so that

$$T^{\omega+1} = -\frac{(\omega+1)q_c}{C}y + \text{const.}$$

Since $T = T_L$ at $y = 0$, the constant of integration is $T_L^{\omega+1}$. The continuum solution is then obtained by setting $y = h$, i.e.

$$q_c = -\frac{C(T_U^{\omega+1} - T_L^{\omega+1})}{(\omega+1)h}. \tag{7.25}$$

For this flow, the free molecule and continuum solutions differ in their functional relationships as well as in the numerical constant. The free molecule heat transfer is proportional to the gas density and is independent of the plate spacing, whereas the continuum heat transfer is inversely proportional to the plate spacing and independent of the density. Therefore,

$$\frac{q_c}{q_f} \propto \frac{1}{\rho h} \propto \frac{\lambda}{h} \propto (Kn), \tag{7.26}$$

so that the ratio of the continuum to the free molecule heat transfer is proportional to the Knudsen number of the flow.

The free molecule heat transfer problem is readily extended to *Couette flow* by giving the upper plate a velocity U in the x-direction (which lies in the plane of the plate), while the lower plate remains stationary. The number fluxes and the normal momentum flux are not affected by this velocity. The shear stress on the lower plate τ_L due to the molecules from the upper plate having a mean velocity U in the x-direction is obtained by setting $s = U\beta$, $\beta = (2RT_U)^{-1/2}$, $\rho = 2n_U m$, and $\theta = \rho/2$ in eqn (4.26). This gives

$$\tau_L = m n_U U (2RT_U/\pi)^{1/2}$$

or, using eqn (7.23),

$$\tau_L = \frac{\rho U}{T_U^{1/2} + T_L^{1/2}}\left(\frac{2RT_U T_L}{\pi}\right)^{1/2}. \tag{7.27}$$

The factor of 2 in the definition of the density was included because the flux equations are applied to the fictitious gas on the opposite side of the upper plate, and this has the number density $2n_U$. The reflected molecules from the lower plate do not contribute to the shear stress so that τ_U is the net value. There will be an equal and opposite shear stress on the upper plate. The energy flux q_U from the upper plate is obtained by making a similar substitution into eqn (4.27), to give

$$q_U = m n_U (2RT_U/\pi)^{1/2}(U^2/2 + 2RT_U). \tag{7.28}$$

In both the above cases, the gas between the two plates is macroscopically uniform. The combination of the two equilibrium distributions gives the following non-equilibrium bimodal distribution function for this gas;

$$f = \frac{1}{\pi^{3/2} n}\left(n_U \beta_U^3 \exp\left[-\beta_U^2\{(u-U)^2 + v^2 + w^2\}\right] + n_L \beta_L^3 \exp\{-\beta_L^2 (u^2 + v^2 + w^2)\}\right). \tag{7.29}$$

The macroscopic properties of the gas are obtained by evaluating the appropriate moments of this distribution function. The stream velocity is in the x-direction with magnitude

$$u_0 = (n_U/n)U, \tag{7.30}$$

and the temperature is

$$T = \{T_L^{1/2}(T_U + \tfrac{1}{3}U^2/R) + T_U^{1/2} T_L\}/(T_U^{1/2} + T_L^{1/2}). \tag{7.31}$$

There is, therefore, a discontinuity in temperature and velocity between the surfaces of the plates and the adjacent gas. Such discontinuities are generally present at surfaces in free molecule flows.

7.4 One-dimensional unsteady flows

The determination of the properties of unsteady flows is best done through the *collisionless Boltzmann equation* using a transformation that was introduced by Narasimha (1962). Since intermolecular collisions are absent, the collision terms on the right-hand side of eqn (3.20) drop out. Then, in the absence of force fields, the collisionless Boltzmann equation is

$$\frac{\partial}{\partial t}(nf) + \boldsymbol{c} \cdot \frac{\partial}{\partial \boldsymbol{r}}(nf) = 0. \tag{7.32}$$

We have seen from eqn (3.6) that nf is identical with the distribution function in phase space $\mathscr{S}$, so that the functional relationship is $nf(\boldsymbol{c},\boldsymbol{r},t)$. We will be concerned with the application of eqn (7.32) to initial value problems, for which

$$nf(\boldsymbol{c},\boldsymbol{r},t=0) = n_i f_i(\boldsymbol{c},\boldsymbol{r}). \tag{7.33}$$

Eqn (7.32) has the form of the Liouville equation and the solution is that nf remains constant along the molecular paths which are the characteristics of the equation, i.e.

$$nf(\boldsymbol{c},\boldsymbol{r},t) = n_i f_i(\boldsymbol{c},\boldsymbol{r}-\boldsymbol{c}t). \tag{7.34}$$

Eqn (7.34) may be multiplied by a molecular quantity Q and integrated over velocity space, to give,

$$n\bar{Q}(\boldsymbol{r},t) = \int_{-\infty}^{\infty} Q n_i f_i(\boldsymbol{c},\boldsymbol{r}-\boldsymbol{c}t)\,\mathrm{d}\boldsymbol{c}.$$

Now make the transformation from **c** to

$$\boldsymbol{r}' = \boldsymbol{r} - \boldsymbol{c}t, \tag{7.35}$$

for which the Jacobian is

$$\frac{\partial(x',y',z')}{\partial(u,v,w)} = -t^3.$$

Therefore,

$$n\bar{Q}(r,t) = \frac{1}{t^3}\int Q n_i f_i\left(\frac{\boldsymbol{r}-\boldsymbol{r}'}{t}, \boldsymbol{r}'\right)\mathrm{d}\boldsymbol{r}'. \tag{7.36}$$

with the limits of the integration being the extent of the gas at time t = 0.

As an example, consider a semi-infinite stationary uniform gas on the negative side of the plane $x = 0$ and separated by a thin wall from a vacuum on the positive side of the plane. At time $t = 0$, the wall is removed and the gas expands freely into the vacuum. Given that the gas is monatomic and originally at the temperature T_1 and number density n_1, the problem is to determine the number density, velocity, and temperature of the gas as a function of x and t, for values of x and t that are very small in comparison with the mean free path and mean collision time, respectively, in the flow. This is called the *free expansion* problem and is probably the simplest one-dimensional unsteady collisionless flow.

The one-dimensional form of eqn (7.36) is

$$n\bar{Q}(x,t) = \frac{1}{t}\int Q n_i f_{u_i}\left(\frac{x-x'}{t}, x'\right)\mathrm{d}x',$$

where (7.37)

$$x' = x - ut.$$

Eqn (4.13) gives the distribution function as a function of $\beta_1 = (2RT_1)^{-1/2}$ and, setting $n_i = n_1$ and $Q = 1$, the density is

$$\begin{aligned} n(x,t) &= \frac{1}{t}\int_{-\infty}^{0} n_1(\beta_1/\pi^{1/2})\exp(-\beta_1^2 u'^2)\,\mathrm{d}x' \\ &= \frac{n_1}{\pi^{1/2}}\int_{\beta_1 x/t}^{\infty} \exp\left(-\frac{\beta_1^2(x-x')^2}{t^2}\right)\mathrm{d}\left(\frac{\beta_1(x-x')}{t}\right). \end{aligned}$$

Therefore, using the standard integrals of Appendix B,

$$\frac{n}{n_1} = \frac{1}{2}\,\mathrm{erfc}\left(\frac{\beta_1 x}{t}\right). \tag{7.38}$$

Similarly, the stream velocity is obtained by setting $Q = u$, to give

$$nu_0 = (\tfrac{1}{2}\pi^{-1/2} n_1/\beta_1)\exp\{-(\beta_1 x/t)^2\}$$

or

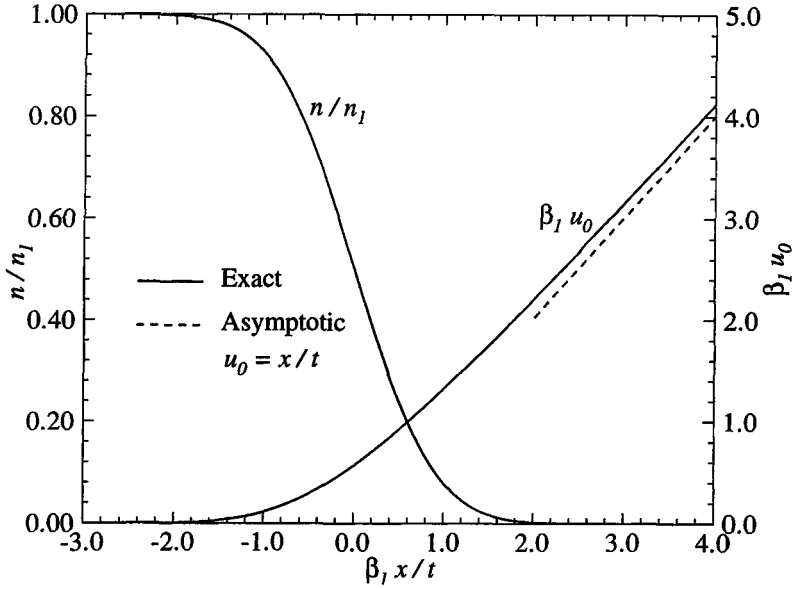


Fig. 7.2 Number density and stream velocity in a free expansion.

$$\beta_1 u_0 = \pi^{-\frac{1}{2}} \exp\{-(\beta_1 x/t)^2\}/\mathrm{erfc}(\beta_1 x/t)\,. \tag{7.39}$$

Note that eqns (7.38) and (7.39) are functions of $\beta_1 x/t$ alone and, at $x=0$, $n/n_1 = \frac{1}{2}$ and $\beta_1 u_0 = \pi^{-\frac{1}{2}}$. The number flux across the plane $x=0$ is therefore constant and equal to $n_1/(2\pi^{\frac{1}{2}}\beta_1)$. A comparison with eqn (7.16) shows that this is identical with the number flux in the steady effusion problem. The behaviour of n/n_1 and $\beta_1 u_0$ as functions of $\beta_1 x/t$ is shown in Fig. 7.2. The density profile is symmetric about the origin, with the perturbation from the undisturbed densities falling as $\exp\{-(\beta_1 x/t)^2\}$.

The analysis is not restricted to the determination of flow fields. For example, consider the extension of the above problem to include a specularly reflecting wall in the plane $x = x_w$. The problem is to determine the pressure p_w on the wall in the free-molecule case in which $(Kn) = \lambda_1/x_w \to \infty$. The pressure on the wall is equal to twice the normal momentum flux at the location $x = x_w$. Therefore, setting $x = x_w$ and $Q = mu^2$ in eqn (7.37),

$$p_w = (2/t)\int_{-\infty}^{0} \pi^{-\frac{1}{2}} \beta_1 n_1 m u^2 \exp(-\beta_1^2 u^2)\,\mathrm{d}x'$$

$$= \frac{2\rho_1\beta_1}{\pi^{\frac{1}{2}}} \int_{\beta_1 x_w/t}^{\infty} \left(\frac{x_w - x'}{t}\right)^2 \exp\left(-\frac{\beta_1^2 (x_w - x')^2}{t^2}\right) \mathrm{d}\left(\frac{x_w - x'}{t}\right),$$

and, again using the standard integrals,

$$\frac{p_w}{p_1} = 1 - \mathrm{erf}\left(\frac{\beta_1 x_w}{t}\right) + \frac{2}{\pi^{1/2}}\left(\frac{\beta_1 x_w}{t}\right)\exp\left(-\frac{\beta_1^2 x_w^2}{t^2}\right). \tag{7.40}$$

The analysis of the free expansion problem may also be extended to cover the expansion of a gas cloud of finite width. If this width is sufficiently small in comparison with the undisturbed mean free path, collisionless analysis is applicable at all locations and at all times. If the gas is initially distributed uniformly between the planes $x = -l$ and $x = +l$, the density and velocity are obtained similarly to eqns (7.38) and (7.39), but with the limits over x' from $-l$ to $+l$, rather than from $-\infty$ to 0. The results for the density and stream velocity are

$$n/n_1 = [\mathrm{erf}\{\beta_1(x+l)/t\} - \mathrm{erf}\{\beta_1(x-l)/t\}]/2 \tag{7.41}$$

and

$$\beta_1 u_0 = \frac{\exp\{-\beta_1^2(x-l)^2/t^2\} - \exp\{-\beta_1^2(x+l)^2/t^2\}}{\pi^{1/2}\,[\mathrm{erf}\{\beta_1(x+l)/t\} - \mathrm{erf}\{\beta_1(x-l)/t\}]}. \tag{7.42}$$

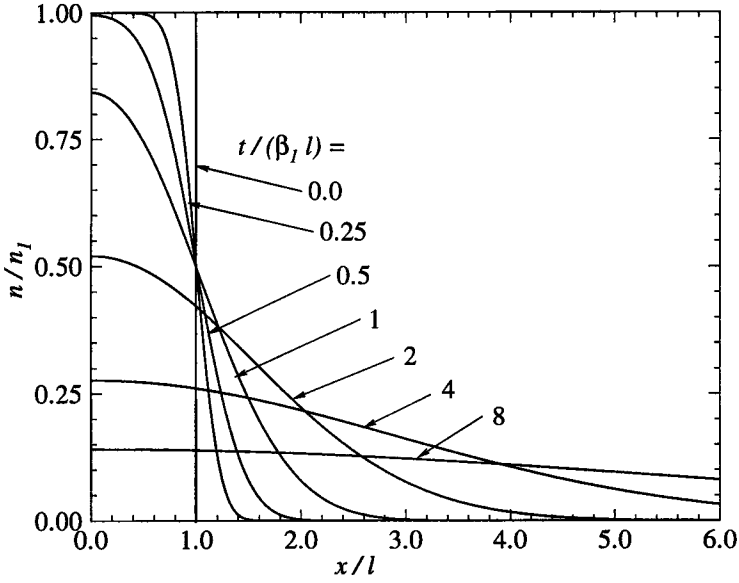


Fig. 7.3 Number density profiles in the expansion of a gas cloud.

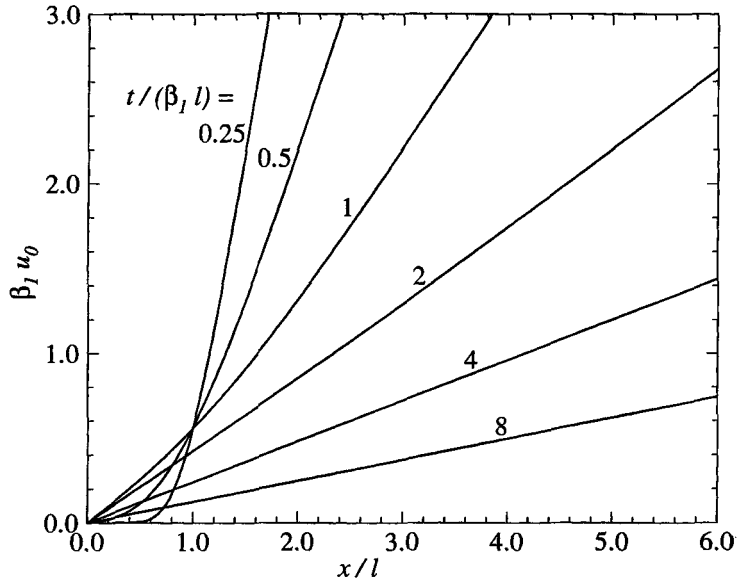


Fig. 7.4 Stream velocity profiles in the unsteady expansion of a gas cloud.

The character of these solutions is shown in Fig. 7.3 for the number density and Fig. 7.4 for the flow velocity. The gas cloud is symmetrical about $x = 0$ at all times. For points within the initial extent of the gas cloud from $x = -l$ to $x = +l$, the density falls continuously, while the velocity increases to a maximum and then decreases. On the other hand, for points that are initially outside the cloud, the maximum velocity is at the leading edge of the cloud, while the density increases to a maximum and then decreases. Narasimha (1962) deals also with the corresponding cylindrically and spherically symmetric flows.

The *one-dimensional piston problem* is another basic flow for which results are required for future reference. A specularly reflecting plane piston impulsively acquires a velocity $\pm U$ at time $t = 0$ and moves either into or away from an initially uniform stationary gas of number density n_1 and temperature T_1. This problem may appear to be quite distinct from the free expansion problem. However, if the problem is viewed from a frame of reference moving with the piston face, the piston may be regarded as a fixed plane at $x = 0$ with the gas on the positive side acquiring the velocity $-U$ towards the plane at time $t = 0$. Similarly, the gas on the negative side of the plane may be regarded as impulsively acquiring a velocity away from the plane. The collisionless Boltzmann equation (7.37) may now be applied to the problem.

The number density becomes

$$n = \frac{1}{t}\int_0^{\infty} n_1 \frac{\beta_1}{\pi^{-\frac{1}{2}}} \exp\{-\beta_1^2(u+U)^2\}\,dx' + \frac{1}{t}\int_{-\infty}^{0} n_1 \frac{\beta_1}{\pi^{-\frac{1}{2}}} \exp\{-\beta_1^2(u-U)^2\}\,dx',$$

the first term corresponding to the gas initially on the positive side of $x = 0$ and the second term to that initially on the negative side. In terms of the standard integrals, this becomes

$$n/n_1 = \pi^{-\frac{1}{2}}\left(\int_{-\beta_1(x/t-U)}^{\infty} \exp\left[-\beta_1^2\{(x-x')/t+U\}^2\right] d\left[\beta_1\{(x-x')/t+U\}\right] \right.$$
$$\left. + \int_{\beta_1(x/t-U)}^{\infty} \exp\left[-\beta_1^2\{(x-x')/t-U\}^2\right] d\left[\beta_1\{(x-x')/t-U\}\right] \right)$$

and may be evaluated to give

$$n/n_1 = 1 + \tfrac{1}{2}\{\mathrm{erf}(\beta_1 x/t + s) - \mathrm{erf}(\beta_1 x/t - s)\}. \tag{7.43}$$

where $s = U\beta$ is the speed ratio of the piston.

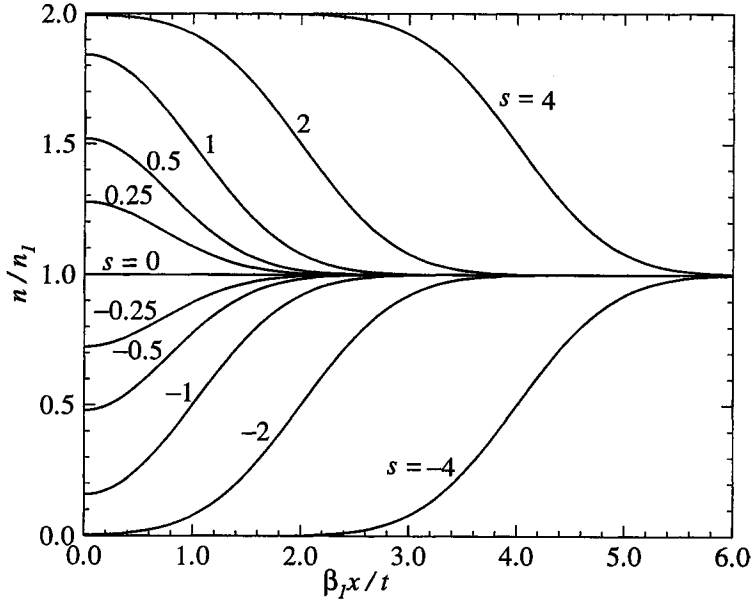


Fig. 7.5 Number density profiles for the one-dimensional piston problem.

The solution for the density is illustrated in Fig 7.5. The sign convention for U has been chosen such that a positive s corresponds to a piston moving into the gas and a negative s to the piston moving away from the gas. For positive values of s, the gas is doubled back with speed $x/t = U$ relative to the piston or plane of symmetry, or $x/t = 2U$ in the frame of reference fixed in the stationary gas into which the piston moves. Superimposed on this, there is a progressive spreading of the front due to the thermal motion of the molecules. The latter effect is dominant at small speed ratios, the number density at the piston face being

$$n/n_1 = 1 + \mathrm{erf}(s)\,. \tag{7.44}$$

For negative values of s, the profiles of n/n_1 lie between 1 and 0 and are symmetrical with those for positive values of s which lie between 1 and 2.

Solutions for the other flowfield properties, such as the stream velocity, may be obtained by substituting the appropriate value of Q into eqn (7.37). The pressure on the piston face could be obtained by a similar calculation to that leading to eqn (7.40) for the free expansion impingement problem. It should be remembered that the pressure tensor is generally anisotropic in a collisionless flow.

7.5 Free-molecule aerodynamics

This is concerned with the pressure, shear stress, and heat transfer at the surfaces of aerodynamic bodies in steady flight. The surface pressure and shear stress may be integrated over the complete body to determine the overall aerodynamic forces on the body. The properties of the flowfield around the body may also be required in some applications. The Knudsen number is generally defined as the ratio of the mean free path in the undisturbed atmosphere to the characteristic dimension of the body, the free-molecule solution being assumed to apply when the Knudsen number is sufficiently high. The justification for this assumption is that, under these circumstances, the molecules reflected from the body generally travel a very large distance before colliding with another molecule. There is then a negligible chance that the products of such collisions will collide with the body, so that the incident molecules are those of the equilibrium free-stream. Note that the most significant collisions in the determination of the validity of the free-molecule assumption are those between the reflected and freestream molecules, and the mean free path for these may be much smaller than the freestream mean free path. This point is particularly important when the speed ratio, and therefore the Mach number, is large and the surface temperature is of the order of the stream temperature.

The surface properties follow from the application of the flux equations of §4.2 to the incident and reflected molecules. The subscripts $_i$ and $_r$ will be used to denote the incident and reflected molecular streams; a flux being regarded as positive if it is directed towards the surface. In the absence of adsorption or emission effects at the surface, the incident number flux to a surface element must be balanced by the reflected number flux. Therefore,

$$\dot{N} = \dot{N}_i + \dot{N}_r = 0; \tag{7.45}$$

$$p = p_i + p_r; \tag{7.46}$$

$$\tau = \tau_i + \tau_r; \tag{7.47}$$

and

$$q = q_i + q_r. \tag{7.48}$$

The results are obviously dependent on the nature of the gas–surface interaction and, in the absence of a general theory for such interactions, calculations will be made for a combination of the classical models of diffuse and specular reflection. From the discussion of these models in §4.5 it can be seen that, for specular reflection,

$$p_r = p_i \quad \text{or} \quad p = 2p_i, \tag{7.49}$$

$$\tau_r = -\tau_i \quad \text{or} \quad \tau = 0, \tag{7.50}$$

and

$$q_r = -q_i \quad \text{or} \quad q = 0. \tag{7.51}$$

The only simplification to eqns (7.45)–(7.48) for diffuse reflection is

$$\tau_r = 0 \quad \text{or} \quad \tau = \tau_i. \tag{7.52}$$

The values of p_i, τ_i, and q_i for small surface elements are given directly by eqns (4.25), (4.26), and (4.31), respectively. These equations provide p_i, τ_i, and q_i as functions of the angle θ between the unit normal vector to the surface and the direction of the freestream velocity U_∞. The *angle of incidence* α of the element is equal to $\pi/2-\theta$, and is the more appropriate angle for aerodynamic studies. The other parameters that are relevant to this study are the freestream density $\rho_\infty = n_\infty m$, the parameter $\beta_\infty = (2RT_\infty)^{-1/2}$ which is a function of the freestream temperature T_∞, and the speed ratio of the freestream $s = U_\infty \beta_\infty$. Eqns (7.49)–(7.52) show that these results provide a complete solution for the specular reflection case.

In diffuse reflection, the molecules are bought to rest relative to the surface and are re-emitted with the equilibrium distribution corresponding to a temperature T_r. The quantities p_r and q_r are therefore given by the flux equations for a stationary gas. From eqn (4.25)

$$p_r = \frac{n_r m}{4\beta_r^2} \tag{7.53}$$

and, from eqn (4.32),

$$q_r = \left(\frac{\gamma+1}{\gamma-1}\right) \frac{n_r m}{8\pi^{1/2}\beta_r^3}. \tag{7.54}$$

The parameter n_r may be regarded as the number density of a fictitious gas

within the surface that is effusing across the surface. The value of n_r is found in §7.1 from the condition, expressed in eqn (7.45), that the net number flux to the surface element is zero. From eqn (7.11)

$$n_r = n_\infty (T_\infty / T_r)^{1/2} [\exp(-s^2 \sin^2\alpha) + \pi^{1/2} s \sin\alpha \{1 + \mathrm{erf}(s \sin\alpha)\}] . \quad (7.55)$$

This fictitious internal number density should not be confused with the number density n_s of the gas just above the surface. Eqn (7.13) gives this as the sum of the fraction of the stream number density that is directed to the surface and the half of n_r that is directed away from the surface, i.e.

$$n_s = \tfrac{1}{2}[\{1 + \mathrm{erf}(s \sin\alpha)\} n_\infty + n_r] . \quad (7.56)$$

The number flux to the surface may be written non-dimensionally as

$$\frac{\dot{N}_i}{n_\infty U_\infty} = \frac{1}{2\pi^{1/2} s} \left[\exp(-s^2 \sin^2\alpha) + \pi^{1/2} s \sin\alpha \{1 + \mathrm{erf}(s \sin\alpha)\}\right], \quad (7.57)$$

and this is illustrated in Fig. 7.6. When s is small in comparison with unity, the number flux ratio is large in comparison with unity. This is because there is a flux to the surface due to the thermal motion of the molecules even when the speed ratio is zero. The number flux is unity to the front of surfaces that are nearly normal to a stream with a high speed ratio. The flux to the rear of such surfaces is extremely small.

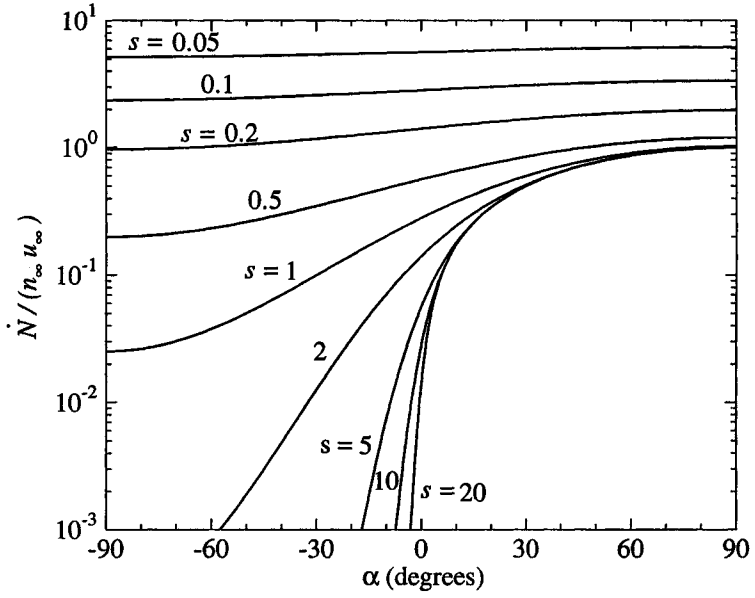


Fig. 7.6 The number flux to a surface element at incidence α.

If a fraction ε of the molecules is reflected specularly and the remaining fraction $1-\varepsilon$ is reflected diffusely, the above results may be combined to obtain general expressions for the pressure, shear stress, and heat transfer. That for the pressure is

$$p/p_\infty = (2\beta_\infty^2 p/\rho_\infty) = \left[(1+\varepsilon)\pi^{-1/2} s\sin\alpha + \tfrac{1}{2}(1-\varepsilon)(T_r/T_\infty)^{1/2}\right] \times \exp(-s^2\sin^2\alpha) + \left[(1+\varepsilon)(\tfrac{1}{2}+s^2\sin^2\alpha) + \tfrac{1}{2}(1-\varepsilon)(T_r/T_\infty)^{-1/2}\pi^{1/2} s\sin\alpha\right]\left[1+\operatorname{erf}(s\sin\alpha)\right]. \quad (7.58)$$

The terms that include the temperature ratio are due to the diffusely reflected molecules. The specularly reflected molecules contribute the same pressure as that due to their incidence on the element. The *pressure coefficient* is related to this pressure ratio by

$$C_p = \frac{p-p_\infty}{\tfrac{1}{2}\rho_\infty U_\infty^2} = \frac{(p/p_\infty)-1}{\tfrac{1}{2}\gamma(Ma)^2} = \frac{(p/p_\infty)-1}{s^2}, \quad (7.59)$$

where the *Mach number* (Ma) is the ratio of the stream speed to the speed of sound. The pressure coefficients can be large in comparison with unity at small speed ratios because the pressure perturbations in collisionless flows are directly proportional to the speed ratio, rather than to the square of the speed ratio as in continuum flows. It is desirable to avoid negative coefficients and, for collisionless flows, it is preferable to define a modified pressure coefficient as

$$C_p' = \frac{p/p_\infty}{s^2}. \quad (7.60)$$

The collisionless pressure is best illustrated by separate plots for the modified pressure coefficients $(C_p')_i$ due to the incident molecules and $(C_p')_r$ due to the reflected molecules. These are shown in Figs 7.7 and 7.8, respectively. The reflected molecule results are for diffuse reflection and the modified pressure coefficient is multiplied by the square root of the ratio of the freestream to the reflected gas temperature. The pressure coefficient due to the incident molecules on a near normal surface in a flow with a very high speed ratio is 2. The pressure due to the reflected molecules is relatively small in high-speed flows unless the surface temperature is very large in comparison with the temperature of the undisturbed freestream. The pressure on surfaces at negative incidences is very small unless the speed ratio is of order unity or less. For fully specular reflection, the pressure due to the reflected molecules would be exactly equal to that due to the incident molecules. The general result for the shear stress is

$$\tau/p_\infty = (2\beta_\infty^2 p)/\rho_\infty = \pi^{-1/2}(1-\varepsilon)s\cos\alpha\left[\exp(-s^2\sin^2\alpha) + \pi^{1/2} s\sin\alpha\,\{1+\operatorname{erf}(s\sin\alpha)\}\right]. \quad (7.61)$$

The *local skin friction coefficient* is defined by

$$c_f = \frac{\tau}{\tfrac{1}{2}\rho_\infty U_\infty} = \frac{\tau/p_\infty}{\tfrac{1}{2}\gamma(Ma)^2} = \frac{\tau/p_\infty}{s^2}. \quad (7.62)$$

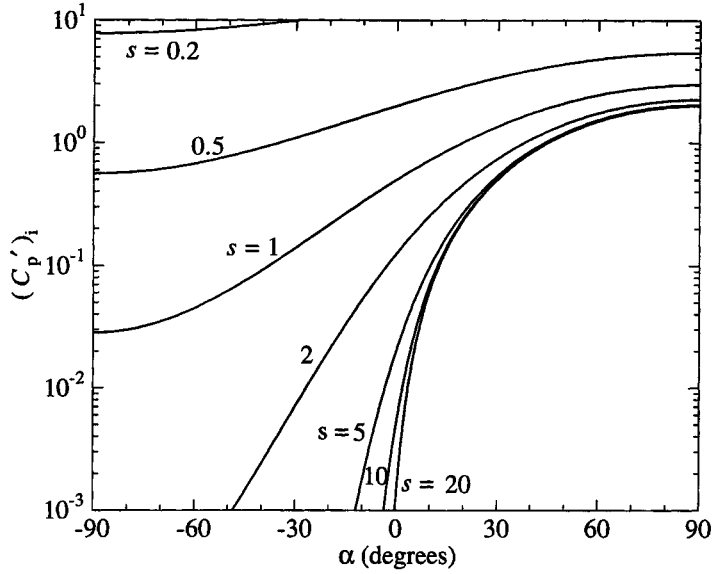


Fig. 7.7 Modified pressure coefficient due to the incident molecules.

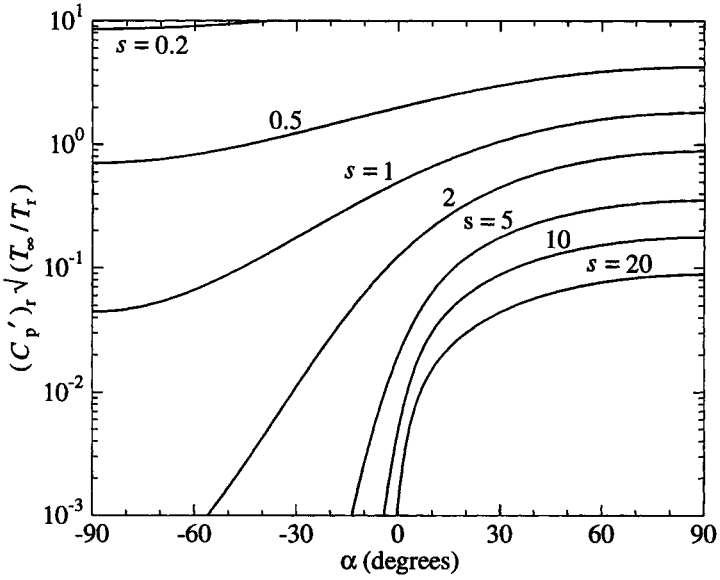


Fig. 7.8 Modified pressure coefficient due to the reflected molecules.

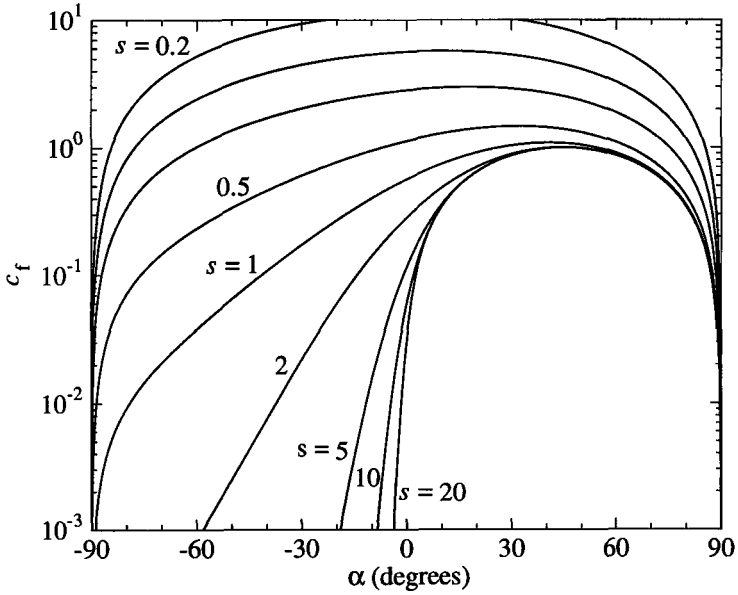


Fig. 7.9 The local skin friction coefficient on a surface element.

The overall shear stress is zero for fully specular reflection and is entirely due to the incident molecules when the reflection is fully diffuse. Figure 7.9 shows that the maximum skin friction coefficient occurs at zero incidence when the speed ratio is extremely small, but is near 45° at very high speed ratios.

The general result for the heat transfer to the surface is

$$2\beta_\infty^3 q/\rho_\infty = (1-\varepsilon)/(2\pi^{1/2})\Big([s^2+\gamma/(\gamma-1)-\{\tfrac{1}{2}(\gamma+1)/(\gamma-1)\}(T_r/T_\infty)] \times\big[\exp(-s^2\sin^2\alpha)+\pi^{1/2}s\sin\alpha\{1+\operatorname{erf}(s\sin\alpha)\}\big] - \tfrac{1}{2}\exp(-s^2\sin^2\alpha)\Big), \tag{7.63}$$

and a *heat transfer coefficient* may be defined by

$$C_h = \frac{q}{\tfrac{1}{2}\rho_\infty U^3} = \frac{2\beta_\infty^3 q/\rho_\infty}{s^3}. \tag{7.64}$$

The term that includes the temperature ratio represents the heat that is carried away by the diffusely reflected molecules. Figures 7.10 and 7.11 illustrate the heat transfer, for $\gamma = 5/3$, through the separate contributions from the incident and reflected molecules. The incident contribution applies also to any fraction of specularly reflected molecules, but this fraction would make a zero contribution to the net heat transfer. For high-speed flows, it can be seen that the reflected heat transfer can equal the incident heat transfer only if the temperature ratio is approximately s^2.

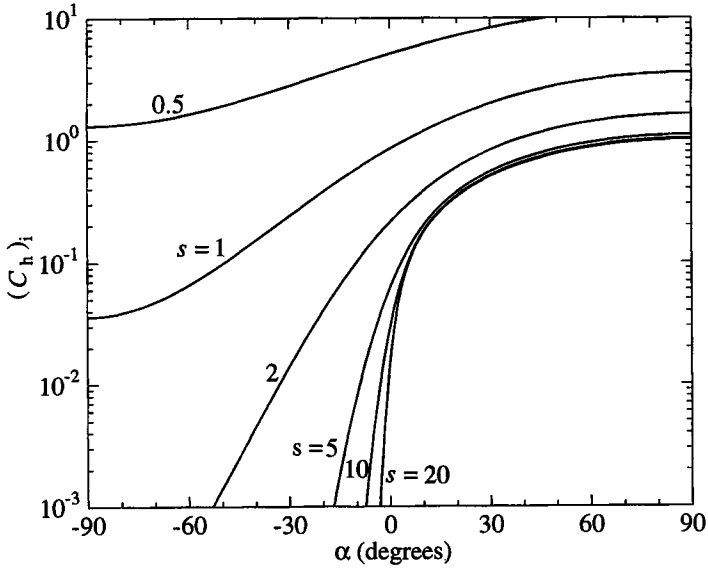


Fig. 7.10 Heat transfer coefficient due to the incident molecules.

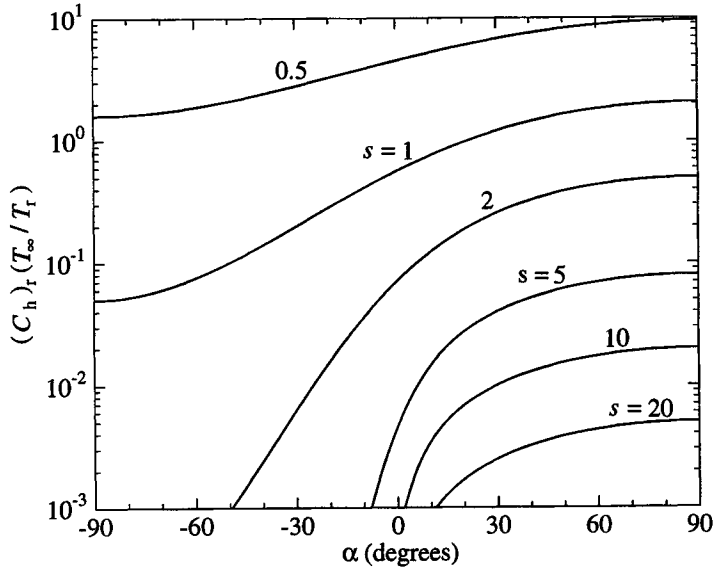


Fig. 7.11 Heat transfer coefficient due to the reflected molecules.

Note that, with the sign convention that has been used in this chapter, the energy accommodation coefficient of eqn (5.54) becomes

$$a_c = \frac{q_i + q_r}{q_i + q_w} = \frac{q}{q_i + q_w}.$$

Therefore, the reflected gas temperature T_r may be replaced by the surface temperature T_w in eqn (7.63) if the right-hand side is multiplied by the accommodation coefficient a_c. For complete thermal accommodation, T_r may be replaced by T_w in all the equations.

The 'adiabatic surface temperature' T_a of an insulated surface element at which $q = 0$ then follows from eqn (7.63) as,

$$\frac{T_a}{T_\infty} = \frac{2(\gamma-1)}{\gamma+1}\left(\frac{\gamma}{\gamma-1} + s^2 - \frac{\tfrac{1}{2}\exp(-s^2\sin^2\alpha)}{\exp(-s^2\sin^2\alpha) + \pi^{1/2} s\sin\alpha\{1+\mathrm{erf}(s\sin\alpha)\}}\right). \quad (7.65)$$

It is instructive to compare the temperature T_a with the continuum stagnation temperature T_0 which is defined by

$$\frac{T_0}{T_\infty} = 1 + \frac{\gamma-1}{2}(Ma)^2 = 1 + \frac{\gamma-1}{\gamma}s^2. \quad (7.66)$$

It can be seen from eqns (7.65) and (7.66) that, for surfaces at positive incidence in a high speed ratio flow, the ratio of the collisionless adiabatic temperature to the continuum stagnation temperature tends to $2\gamma/(\gamma+1)$. This ratio is 5/4 for a monatomic gas, 7/6 for a diatomic gas, and so on. The general behaviour of this temperature ratio is shown in Fig. 7.12. The ratio falls below unity for surfaces at negative incidence and actually tends to zero at the rear of a vertical plate in a hypersonic flow. This is because the molecules that reach such a surface have a very small velocity relative to the surface. However, the number flux under these conditions is so minute that the result has little significance.

The above equations may be applied directly to determine the aerodynamic forces on a thin flat plate at incidence α in a stream of speed ratio s. The equations apply to the lower surface for positive values of α and to the upper surface for negative values of α. The upper surface pressure may be subtracted from the lower surface pressure to obtain the net force per unit area, and the normal force coefficient can be written from eqn (7.58) as

$$C_N = \left[2(1+\varepsilon)\pi^{-1/2} s\sin\alpha\exp(-s^2\sin^2\alpha) + (1-\varepsilon)(T_r/T_\infty)^{1/2}\pi^{1/2} s\sin\alpha + (1+\varepsilon)(1+2s^2\sin^2\alpha)\mathrm{erf}(s\sin\alpha)\right]/s^2. \quad (7.67)$$

The parallel force coefficient can be obtained similarly from the shear stress eqn (7.61) as

$$C_P = 2(1-\varepsilon)\pi^{-1/2} s\cos\alpha\{\exp(-s^2\sin^2\alpha) + \pi^{1/2} s\sin\alpha\,\mathrm{erf}(s\sin\alpha)\}/s^2. \quad (7.68)$$

The normal and parallel force coefficients may now be resolved into the directions normal to and parallel to the stream direction in order to obtain the lift and drag coefficients for the flat plate. The reference area is again based on the planform area, and the results are

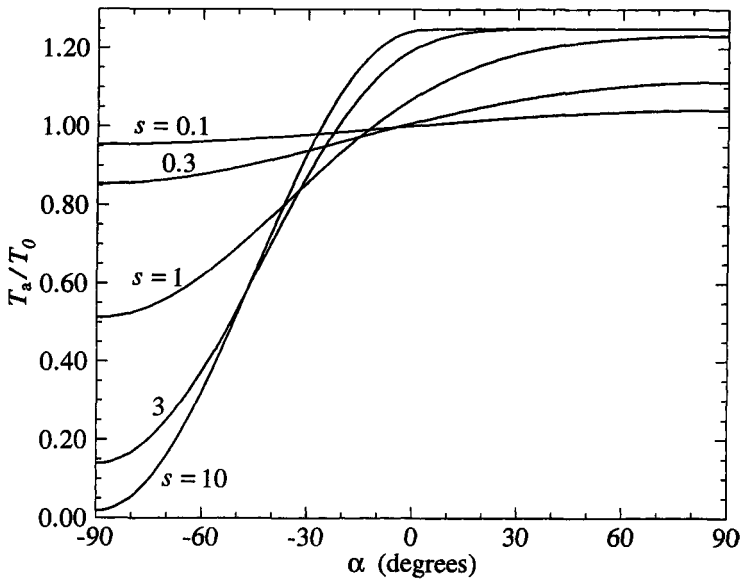


Fig. 7.12 The ratio of the collisionless adiabatic surface temperature to the continuum stagnation temperature in a monatomic gas.

$$C_L = \frac{4\varepsilon}{\pi^{1/2}s}\sin\alpha\cos\alpha\exp(-s^2\sin^2\alpha) + \frac{\cos\alpha}{s^2}\{1+\varepsilon(1+4s^2\sin^2\alpha)\} \times \operatorname{erf}(s\sin\alpha) + \frac{(1-\varepsilon)}{s}\pi^{1/2}\sin\alpha\cos\alpha\left(\frac{T_r}{T_\infty}\right)^{1/2} \tag{7.69}$$

and

$$C_D = 2\frac{\{1-\varepsilon\cos(2\alpha)\}}{\pi^{1/2}s}\exp(-s^2\sin^2\alpha) + \frac{\sin\alpha}{s^2}[1+2s^2+\varepsilon\{1-2s^2\cos(2\alpha)\}]\operatorname{erf}(s\sin\alpha) + \frac{(1-\varepsilon)}{s}\pi^{1/2}\sin^2\alpha\left(\frac{T_r}{T_\infty}\right)^{1/2}. \tag{7.70}$$

The lift coefficient and lift–drag ratio of a flat plate at a speed ratio of 10 and with the surface temperature equal to the freestream temperature are shown in Figs 7.13 and 7.14, respectively. These conditions are representative of orbital conditions, and show that lifting performance in free-molecule flow is poor for fully diffuse flow. Good lifting performance requires the assumption of an unrealistic degree of specular reflection. Lifting performance is slightly better at lower speed ratios.

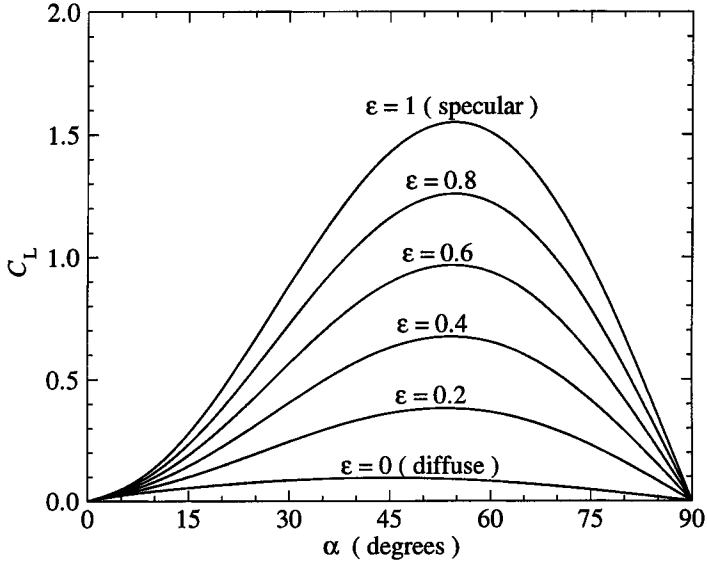


Fig. 7.13 Lift of a flat plate at $s = 10$ and $T_r = T_\infty$.

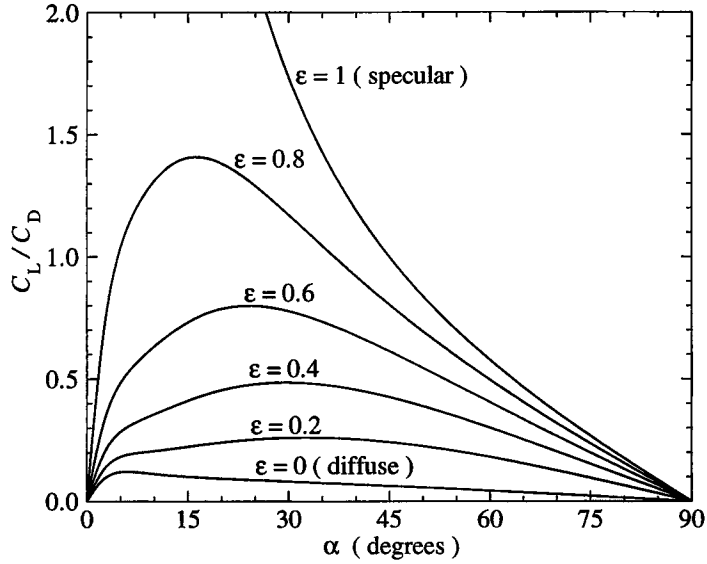


Fig. 7.14 Lift–drag ratio of a flat plate at $s = 10$ and $T_r = T_\infty$.

For bodies other than flat plates, eqns (7.58), (7.61), and (7.63) must be integrated over the surface. This has been done for many simple geometries and a list of the then available solutions was given by Schaaf and Chambre (1961). The chances of obtaining a closed form analytical solution depend on the geometrical complexity of the problem.

The procedure may be illustrated through the calculation of the drag coefficient of a sphere of radius r. Polar coordinates are chosen such that the pole points in the opposite direction to the stream. The polar angle θ is then identical to the angle θ in the flux equations of §4.2 and, in terms of the angle of incidence α that has been used in this section, it is equal to $\pi/2 - \alpha$. The drag coefficient based on the frontal area of the sphere is

$$C_D = \frac{D}{\tfrac{1}{2}\rho_\infty U^2 \pi r^2} = \frac{\int_0^\pi (p\cos\theta + \tau\sin\theta)2\pi r^2 \sin\theta \, d\theta}{\tfrac{1}{2}\rho_\infty U^2 \pi r^2}.$$

Eqn (7.58) for the pressure and eqn (7.61) for the shear stress may be substituted into this equation and the integral evaluated to give,

$$C_D = \frac{2s^2+1}{\pi^{\frac{1}{2}} s^3}\exp(-s^2) + \frac{4s^4+4s^2-1}{2s^4}\operatorname{erf}(s) + \frac{2(1-\varepsilon)\pi^{\frac{1}{2}}}{3s}\left(\frac{T_w}{T_\infty}\right)^{\frac{1}{2}}. \quad (7.71)$$

A remarkable feature of eqn (7.71) is that the fraction of specular reflection disappears from all terms except the term including the temperature ratio that is due to the reflected molecules. Therefore, for a cold sphere, the drag coefficient is the same for both diffuse and specular reflection, with the common value tending to two as the speed ratio becomes large. This is an accident of geometry in that the high pressure near the stagnation point under specular conditions is matched by the high shear stress near the maximum radius under diffuse conditions. This result should not be interpreted as meaning that the drag coefficient of a sphere is independent of the gas–surface interaction model. It was pointed out in §5.6 that a combination of specular and diffuse reflection is only an approximation to the real scattering law, and some models lead to a limiting drag coefficient quite different from two.

The drag of some shapes can be highly dependent on the model for the gas–surface interaction. For example, eqn (7.70) shows that the drag coefficient of a cold vertical flat plate at a high speed ratio is

$$C_D = 2 + 2\varepsilon. \quad (7.72)$$

The factor of two difference between the diffuse and specular limits occurs in this case because there is no contribution from the shear stress, the drag being due entirely to the pressure. For a flat plate at zero incidence, there is a finite drag due to the shear stress in the diffuse reflection case, but the drag is zero for the unrealistic case of completely specular reflection.

In the case of a gas mixture, all of the above equations may be applied separately to each molecular species, and the results may then be superimposed.

7.6 Thermophoresis

A thermal force acts on a small suspended particle in a gas with a temperature gradient, even though the temperature of the particle may be uniform and equal to the local gas temperature. The force is essentially due to the asymmetry of the Chapman–Enskog distribution function and the phenomenon is called *thermophoresis*. At sufficiently low densities or for sufficiently small particles, intermolecular collisions may be neglected and the magnitude of the effect is predicted by a free-molecule analysis. In addition to the non-equilibrium boundary conditions, this example differs from the earlier collisionless flows in that the corresponding continuum flow is not well defined.

Consider a spherical particle of radius r and, as shown in Fig. 7.15, take the origin at the centre of the sphere and choose the x-axis in the direction of the temperature gradient. The first approximation to the Chapman–Enskog distribution is then given by eqn (3.85) as

$$f = f_0\left\{1 - \tfrac{4}{5}\frac{\beta^2 K}{RT}\left(\beta^2 c^2 - \tfrac{5}{2}\right)u\,\frac{\mathrm{d}T}{\mathrm{d}x}\right\}. \tag{7.73}$$

Now consider the small surface element at point P on the sphere defined by the polar angle θ and azimuth angle ϕ. Axes x', y', z' are taken with x' normal to and directed towards the surface element and y' along the line of intersection of the plane defined by OPx and the plane of the surface element. Eqns (1.18) and (3.3) may then be applied in the x', y', z' frame of reference, with f from eqn (7.73), to give the flux of the quantity Q to the surface element as

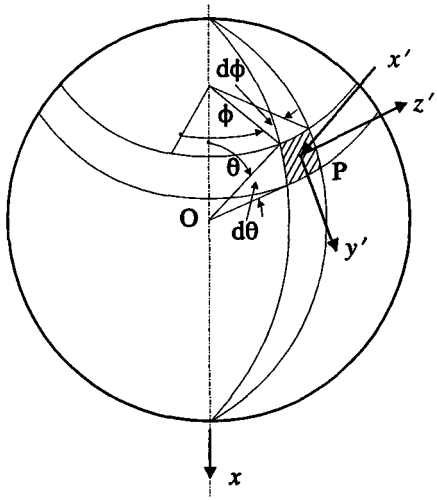


Fig. 7.15 Coordinate system for the analysis of the thermophoretic force on a sphere.

$$\frac{n\beta^3}{\pi^{3/2}}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{\infty} QU\exp\{-\beta^2(U^2+V^2+W^2)\}$$
$$\times\left\{1-\frac{4}{5}\frac{\beta^2 K}{\rho RT}\left(\beta^2(U^2+V^2+W^2)-\frac{5}{2}\right)(U\cos\theta+V\sin\theta)\frac{dT}{dx}\right\}dU\,dV\,dW,$$

where U, V, and W are the velocity components in the x', y', and z' directions, respectively. By putting $Q=mU$ and $Q=mV$ we obtain, respectively, the pressure p_i and shear stress τ_i due to the impact of the incident molecules. The results are

$$p_i=\frac{\rho}{4\beta^2}-\frac{2\beta K}{5\pi^{1/2}}\cos\theta\frac{dT}{dx}, \tag{7.74}$$

and

$$\tau_i=\frac{\beta K}{5\pi^{1/2}}\sin\theta\frac{dT}{dx}. \tag{7.75}$$

A term, such as the first one on the right-hand side of eqn (7.74), that is uniform around the sphere obviously makes no contribution to the net force on the sphere. A property of the Chapman–Enskog distribution is that the number flux is isotropic so that, if the surface temperature is uniform, the force exerted by diffusely reflected molecules is also uniform and makes no net contribution to the force. The pressure due to the specularly reflected molecules is equal to p_i. The shear stress due to diffusely reflected molecules is zero and that due to any specularly reflected molecules is equal to $-\tau_i$. The net force is therefore given by

$$F=-\frac{\beta K}{5\pi^{1/2}}\frac{dT}{dx}\int_0^{\pi}\{2(1+\varepsilon)\cos^2\theta+(1-\varepsilon)\sin^2\theta\}\,2\pi r^2\sin\theta\,d\theta,$$

where ε is the fraction of specularly reflected molecules. Therefore,

$$F=-\frac{16\pi^{1/2}}{15}\beta r^2 K\frac{dT}{dx}. \tag{7.76}$$

This result was first obtained by Waldmann (1959). Note that the disappearance of the fraction ε of specular reflection from the final result is due to the spherical geometry and would not occur for particles which have a different shape. This case with a uniform temperature over the surface is appropriate to a particle with a very high thermal conductivity. A temperature gradient would be induced in a particle that has a low thermal conductivity, and this would increase the force. A force due to the unbalance in the pressure due to diffusely reflected molecules acts on any particle that has a non-uniform temperature, even if the surrounding gas has a uniform temperature. The non-uniform surface temperature is most commonly caused by thermal radiation, and the phenomenon in an equilibrium background gas is called *photophoresis*.

7.7 Flows with multiple reflection

The analysis of collisionless flows becomes much more difficult if the molecules that are reflected from a surface may re-impinge on the surface instead of escaping completely. Flows that involve multiple interactions have practical application to internal flows and also to external flows past bodies of complex geometry.

Consider a surface exposed to a gas under free-molecule conditions. The number of molecules incident per unit time on a surface element $\mathrm{d}S$ at the location S directly from the external gas may be denoted by $\dot{N}_1(S)\mathrm{d}S$. Now let $P(S',S)$ be the probability that a molecule reflected from an element $\mathrm{d}S'$ at location S' strikes location S. The number of molecules per second which strikes $\mathrm{d}S$ as their second collision with the surface is, therefore,

$$\dot{N}_2(S)\mathrm{d}S = \int_S P(S',S)\dot{N}_1(S')\,\mathrm{d}S'\,\mathrm{d}S.$$

Similarly, the number striking $\mathrm{d}S$ as their third collision is

$$\dot{N}_3(S)\mathrm{d}S = \int_S P(S',S)\dot{N}_2(S')\,\mathrm{d}S'\,\mathrm{d}S.$$

Therefore, if $\dot{N}(S)$ is the total number flux at location S,

$$\begin{aligned}\dot{N}(S) &= \dot{N}_1(S) + \dot{N}_2(S) + \dot{N}_3(S) + \ldots \\ &= \dot{N}_1(S) + \int_S P(S',S)\{\dot{N}_1(S') + \dot{N}_2(S') + \dot{N}_3(s') + \ldots\}\,\mathrm{d}S',\end{aligned}$$

or

$$\dot{N}(S) = \dot{N}_1(S) + \int_S P(S',S)\dot{N}(S)\,\mathrm{d}S'. \tag{7.77}$$

This is a Fredholm integral equation of the second kind and its solution must form part of the analysis of collisionless flows that involve multiple reflections.

As an example, consider the free-molecular flow through a circular tube of length b and radius r. The centreline of the tube lies along the x-axis and the origin is at the entrance to the tube. The molecules that pass through the tube will be made up of those that pass directly through it without any collision with the inside surface, plus a fraction of those emitted from each element of the inside surface. The total flux $\dot{N}_\mathrm{t}$ through the tube may therefore be written

$$\dot{N}_\mathrm{t} = \dot{N}_\mathrm{d} + \int_0^b \dot{N}(x)P_\mathrm{e}(x)\,\mathrm{d}x, \tag{7.78}$$

where $\dot{N}_\mathrm{d}$ is the direct number flux, $P_\mathrm{e}(x)$ is the probability that a molecule reflected from the element of length $\mathrm{d}x$ at x passes through the tube, and $\dot{N}(x)$ follows from the axially symmetric version of eqn (7.77), i.e.

$$\dot{N}(x) = \dot{N}_1(x) + \int_0^b P(x',x)\,\dot{N}(x)\,\mathrm{d}x. \tag{7.79}$$

Solutions of this problem were first obtained by Clausing (1932) and have been discussed in detail by Patterson (1971). The functions $\dot{N}_d$, $P_e(x)$, $\dot{N}_1(x)$, and $P(x',x)$ may be found for this simple geometry by straightforward but tedious analysis. However, the solution of the integral equation for $\dot{N}(x)$ must be carried out numerically. The final integration of eqn (7.78) requires the assumption that $\dot{N}(x)$ is a linear function of x, with the coefficients provided by the numerical solution of eqn (7.79).

The above problem is of comparatively simple geometry and the predominantly analytical approach through the integral equation becomes quite unworkable for more complex problems. On the other hand, this class of problem is ideally suited to a probabilistic numerical method called the *test-particle Monte Carlo method* (Davis 1960). A digital computer is used to calculate a very large number of typical molecular trajectories, and these collectively predict the behaviour of the real system. Since intermolecular collisions may be neglected, these trajectories are independent of one-another and they may be generated serially. The method will now be illustrated through its application to the circular tube flux problem to calculate the numbers N_t and N_d that are associated with the fluxes.

The flow chart for the simulation program is shown in Fig. 7.16. The additional notation that is used in this chart is as follows: $L = b/r$ is the length-radius ratio of the tube, N_e is the total number of test particles that enter the tube, r_1 is the initial radius of the particle at $x = 0$, l_1, m_1, and n_1 are the direction cosines of the initial trajectory with the x-, y-, and z-axes, x_c is the x-coordinate of the intersection point either with the inside of the cylinder or its imaginary projections outside $x = 0$ to b, and l_c, m_c, and n_c are the direction cosines of the particle trajectory after diffuse reflection from the inside surface of the cylinder.

The distribution function for the initial radius is one of those dealt with in Appendix C, and the required expression follows directly from eqn (C6) as

$$r_1 = (R_f)^{1/2}\, r. \tag{7.80}$$

This chooses the initial radii at random from a uniform distribution over the entry plane. However, the number of trajectories is extremely small in comparison with the number of molecules that would be expected to enter the tube in a real gas, and the variance due to the scatter in the initial state may be avoided by setting a uniform distribution. The equation for the initial radius of the Nth trajectory is then

$$r_1 = \{(N - 0.5)/N_1\}^{1/2}\, r. \tag{7.81}$$

The number flux of molecules effusing across the entry plane of the tube is given by eqn (4.20) with $Q = 1$. The external gas is in equilibrium and the distribution function may be put into the polar coordinate form of eqn (4.4)

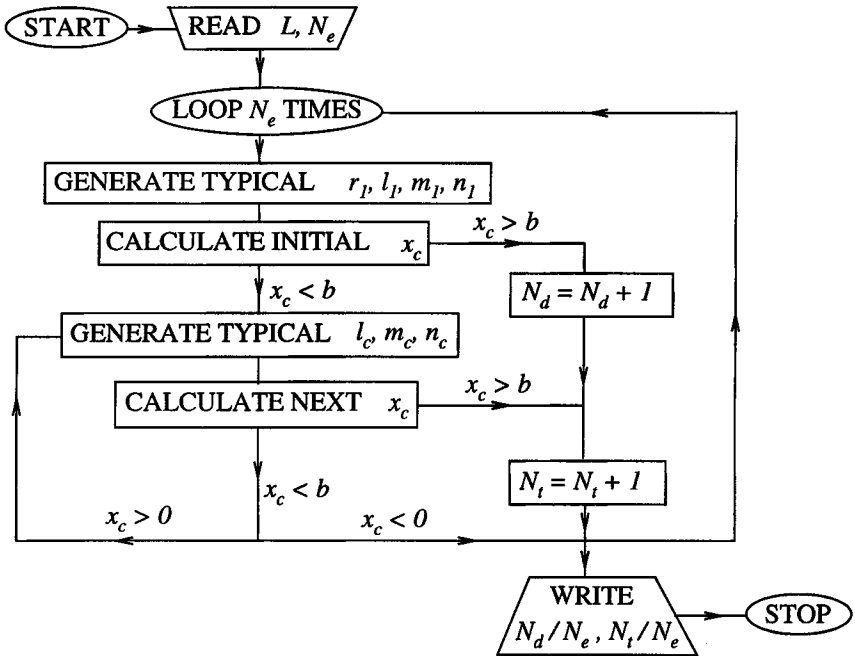


Fig. 7.16 Flow chart of the test-particle Monte Carlo program for the cylindrical tube flux program.

with the polar direction along the axis of the tube. The u velocity component may then be written $c'\cos\theta$ and the number flux of molecules is

$$\dot{N}_i = \frac{n\beta^3}{\pi^{3/2}} \int_0^\infty \int_0^{2\pi} \int_0^{\pi/2} c'^3 \exp(\beta^2 c'^2) \sin\theta\cos\theta \, d\theta \, d\phi \, dc' . \tag{7.82}$$

The probability that a molecule passes through the tube depends on its initial direction at the entry plane and on the direction in which it is reflected at any intersections with the inside of the tube. In particular, this probability is independent of the molecular speed c' and there is no need to consider the distribution of this quantity.

The azimuth angle ϕ is uniformly distributed between 0 and 2π and a typical value is, from eqn (C5),

$$\phi = 2\pi R_f. \tag{7.83}$$

The distribution of the polar angle θ is best dealt with through the distribution of $\cos\theta$ because

$$\sin\theta\cos\theta \, d\theta = \cos\theta \, d(\cos\theta) .$$

The distribution function for $\cos\theta$ is therefore identical with that for the radius r which led, through eqn (C6), to eqn (7.79). The limits on $\cos\theta$ are from 0 to 1 and we can write

$$\cos\theta = (R_f)^{1/2}. \tag{7.84}$$

The direction cosines l_1, m_1, and n_1 are given by

$$l_1 = \cos\theta,$$

$$m_1 = \sin\theta\cos\phi,$$

and (7.85)

$$n_1 = \sin\theta\sin\phi.$$

The x-coordinate of the intersection of the particle trajectory with the cylinder, or its projections outside $x=0$ to b, is readily obtained from elementary three-dimensional coordinate geometry. A general theorem with wide application to simulation studies is that the intersection of the line

$$x = x_i + l_1 s$$

$$y = y_i + m_1 s \tag{7.86}$$

$$z = z_i + n_1 s$$

and the quadric surface

$$f(x,y,z) \equiv a_{11}x^2 + a_{22}y^2 + a_{33}z^2 + 2a_{23}yz + 2a_{31}zx + 2a_{12}xy + 2a_{14}x + 2a_{24}y + 2a_{34}z + a_{44} = 0 \tag{7.87}$$

is given by the roots of

$$A_1 s^2 + 2A_2 s + A_3 = 0, \tag{7.88}$$

where

$$A_1 = a_{11}l_1^2 + a_{22}m_1^2 + a_{33}n_1^2 + 2a_{23}m_1n_1 + 2a_{31}n_1l_1 + 2a_{12}l_1m_1,$$

$$A_2 = l_1(a_{11}x_i + a_{12}y_i + a_{13}z_i + a_{14}) + m_1(a_{21}x_i + a_{22}y_i + a_{23}z_i + a_{24}) + n_1(a_{31}x_i + a_{32}y_i + a_{33}z_i + a_{34}),$$

and

$$A_3 = f(x_i, y_i, z_i).$$

Real roots of eqn (7.88) may be substituted into eqn (7.86) to determine the points of intersection. Since the present case is axially symmetric, we may choose $y_i = -r_1$ and $z_i = 0$. The coefficients a_{22} and a_{33} are equal to unity, $a_{44} = -r^2$, and all the other coefficients are zero. Eqns (7.86)–(7.88) then give the following equation for the x-coordinate of the intersection of the particle trajectory with the cylinder

$$x_c = l_1[r_1m_1 + \{r^2(m_1^2 + n_1^2) - r_1^2n_1^2\}^{1/2}]/(m_1^2 + n_1^2). \tag{7.89}$$

Because of the axial symmetry, the y-and z-coordinates of the point of intersection may be set to $-r$ and 0, respectively. If the point of intersection lies within the cylinder, the assumption of diffuse reflection means that the selection of the direction cosines l_c, m_c, and n_c is similar to the above selection of m_1, l_1, and n_1, respectively. Note that the flux direction is now along the y-, rather than the x-, axis. The x-coordinate of the subsequent point of intersection with the cylinder is, again using eqns (7.86)–(7.89),

$$x = x - c + 2l_c r m_c/(m_c^2 + n_c^2). \tag{7.90}$$

This value of x becomes the next value of x_c.

A FORTRAN 77 implementation of this method is listed in Appendix D. This is a very simple program that contains only about 30 executable statements. The value of the ratio N_t/N_e that was calculated by Clausing (1932) for $L = 1$ was 0.672. A personal computer with a 33 MHz i486 CPU calculates 10,000 trajectories per second for this case. The results from 10^7 typical trajectories were $N_t/N_e = 0.6720$, while $N_d/N_e = 0.3820$. The expected statistical scatter for a sample of 10^7 is much less than one part in 1000, so that this level of agreement is to be expected. The scatter is more clearly demonstrated by calculating the 10^7 trajectories as 10,000 runs, each with a sample of 1000. The results are shown in Fig. 7.17.

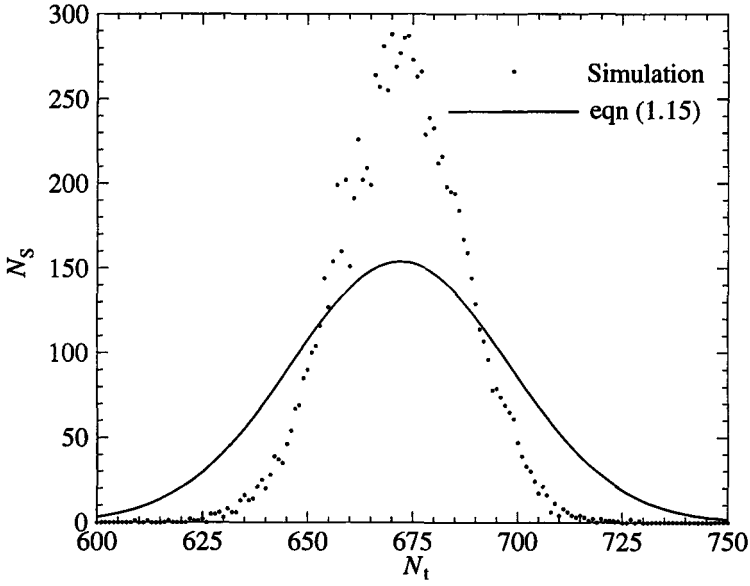


Fig. 7.17 A comparison of the scatter associated with 10,000 separate Monte Carlo simulations with the normal distribution of eqn (1.15).

N_S is the number of trajectories that have a particular value of N_t. The calculated distribution is compared with the normal distribution that is defined by eqn (1.15). For a total sample of 1000, the normally distributed value of N_S may be written as a function of N_t, i.e.

$$N_S = 10000(2\pi\bar{N}_t)^{-\frac{1}{2}} \exp\{-(N_t - \bar{N}_t)^2/(2\bar{N}_t)\}, \tag{7.91}$$

where the mean value $\bar{N}_t$ is here equal to 672. This has a standard deviation equal to the square root of the mean value, or approximately 26. The actual scatter is less than that predicted by the normal distribution, and almost all the results lie within 26 of the mean value. If the results are fitted to a normal distribution, the actual standard deviation is approximately 14. A useful way of estimating the expected magnitude of the scatter is to divide the result by the square root of the sample size. For this case, the estimate of the standard deviation is approximately 21.

As noted earlier, the number of molecules that are involved in a real flow is generally many orders of magnitude larger than the sample size in the simulation, and it is desirable to minimize the scatter. The scatter in the initial radius of the molecule at the entry plane may be reduced by using eqn (7.81), but this leads to an almost imperceptible reduction in the scatter. However, the direction cosines of the entry molecules may also

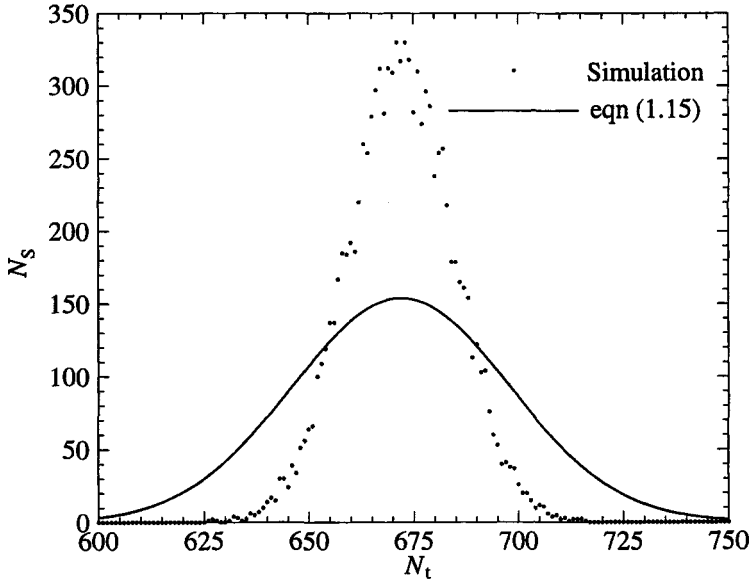


Fig. 7.18 Similar to Fig. 7.17, but with variance reduction applied to the initial values.

be distributed more uniformly. The program requires three random fractions while setting the initial conditions of each trajectory, so that the initial distribution is three-dimensional. The loop of N over NTR in program FMF (Appendix D) may be replaced by three nested loops of N1, N2, and N3 over the cube root of NTR, say NTC. The first random fraction can be replaced, for the N1th trajectory, by

$$(\mathrm{N1} - R_f)/\mathrm{NTC}, \tag{7.92}$$

with similar expressions involving N2 and N3 for the other random fractions. This ensures that the initial states of the trajectories are distributed more uniformly over the three-dimensional space. This is a form of *variance reduction* and the resulting scatter is shown in Fig. 7.18. The standard deviation of a normal distribution fitted to the actual scatter is approximately 12. The scatter is inversely proportional to the square root of the sample size, so that a similar reduction in the scatter could be obtained by an increase in the sample of one third. If the molecular speed had been required for the simulation, the initial distribution would have been four-dimensional, and the variance reduction procedure would be barely practical for a sample of 1000 in each run.

Similar agreement with the (largely) analytical solution of Clausing has been obtained for other values of the length–radius ratio. The implementation of the test-particle Monte Carlo method is much easier than the implementation of the more analytical method based on the Fredholm integral equation. The cost of the computation of 10^7 or 10^8 trajectories is negligible, so that four figure accuracy is readily obtained with the Monte Carlo method. However, the major advantage of the Monte Carlo method is the ease with which it is able to deal with more complex flow geometries. Changes that would make the integral equation approach quite impractical can generally be made through alterations to several statements in the simulation program. A number of configurations of engineering interest were dealt with by Davis (1961).

The above example required only the determination of the ratio of the exit to the entry number fluxes. When applying the Monte Carlo approach to problems such as the determination of the drag coefficient of complex shapes, it becomes necessary to relate the momentum transfer to the gas density. The simplest procedure for steady flows is to assume that each simulated trajectory represents a much larger number of real trajectories such that the computed trajectories represent those occurring in the real gas over unit time. The sum of the momentum and energy transfer over all trajectories then gives the momentum and energy fluxes, and the pressure, shear stress, and heat transfer follow.

The test-particle Monte Carlo method is not suited to the computation of unsteady flows. For these it is preferable to adopt a collisionless version of the direct simulation Monte Carlo (or DSMC) method that is described in Chapter 10. In this method, a large number of trajectories are computed simultaneously, and there is a computational time parameter that corresponds to physical time in the real gas.

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